

Instructions. (15 points) Section 12.5. Show relevant work for full credit.

- (5pts) 1. Determine whether the lines L_1 and L_2 intersect. If they do, find the point of intersection. If they do not, determine whether they are parallel or skew.

$$L_1 : x = 1 + t, \quad y = 2 + 2t, \quad z = -1 + 3t$$

$$L_2 : x = 3 + s, \quad y = 6 - s, \quad z = 5 + s$$

Solution:

Set components equal:

$$1 + t = 3 + s \quad \implies \quad t - s = 2 \quad (\text{i})$$

$$2 + 2t = 6 - s \quad \implies \quad 2t + s = 4 \quad (\text{ii})$$

$$-1 + 3t = 5 + s \quad \implies \quad 3t - s = 6 \quad (\text{iii})$$

From (i) + (ii): $3t = 6$, so $t = 2$ and $s = 0$.

Check (iii): $3(2) - 0 = 6 \checkmark$

All three equations are satisfied, so the lines **intersect** at

$$(1 + 2, 2 + 4, -1 + 6) = \boxed{(3, 6, 5)}.$$

- (5pts) 2. Find the distance from the point $P(3, 1, -2)$ to the plane $2x - 2y + z = 6$.

Solution:

The plane has normal vector $\mathbf{n} = \langle 2, -2, 1 \rangle$.

Method 1 (normal line). The line through P perpendicular to the plane is

$$x = 3 + 2t, \quad y = 1 - 2t, \quad z = -2 + t.$$

Substitute into $2x - 2y + z = 6$:

$$2(3 + 2t) - 2(1 - 2t) + (-2 + t) = 6 \implies 9t + 2 = 6 \implies t = \frac{4}{9}.$$

The foot of the perpendicular is $F = \left(\frac{35}{9}, \frac{1}{9}, \frac{-14}{9}\right)$.

$$d = |PF| = \sqrt{\left(\frac{8}{9}\right)^2 + \left(\frac{-8}{9}\right)^2 + \left(\frac{4}{9}\right)^2} = \frac{\sqrt{144}}{9} = \boxed{\frac{4}{3}}.$$

Method 2 (scalar projection). Choose any point on the plane, e.g. $Q(0, 0, 6)$.

$$\overrightarrow{QP} = \langle 3, 1, -8 \rangle, \quad |\mathbf{n}| = \sqrt{4 + 4 + 1} = 3.$$

The distance is the absolute scalar projection of $\overrightarrow{QP}$ onto $\mathbf{n}$:

$$d = \frac{|\overrightarrow{QP} \cdot \mathbf{n}|}{|\mathbf{n}|} = \frac{|6 - 2 - 8|}{3} = \boxed{\frac{4}{3}}.$$

- (5^{pts}) 3. Find a vector equation for the line of intersection of the planes

$$x + y - z = 2 \quad \text{and} \quad 2x - y + z = 1.$$

Solution:

Direction vector. The line lies in both planes, so its direction is perpendicular to both normal vectors. With $\mathbf{n}_1 = \langle 1, 1, -1 \rangle$ and $\mathbf{n}_2 = \langle 2, -1, 1 \rangle$:

$$\mathbf{n}_1 \times \mathbf{n}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & -1 \\ 2 & -1 & 1 \end{vmatrix} = \langle 0, -3, -3 \rangle.$$

We can use $\mathbf{d} = \langle 0, 1, 1 \rangle$ (dividing by -3).

Point on the line. Set $z = 0$ and solve:

$$\begin{aligned} x + y &= 2 \\ 2x - y &= 1 \end{aligned}$$

Adding: $3x = 3$, so $x = 1$, $y = 1$. A point on the line is $(1, 1, 0)$.

Vector equation:

$$\mathbf{r}(t) = \langle 1, 1, 0 \rangle + t \langle 0, 1, 1 \rangle.$$